Optimising attention on Apple M1

At $N = 4096, d = 64$, naive attention takes 1836.944 ms. The best single-threaded variant takes 80.528 ms (22.81× faster), and 8 OpenMP threads take 26.335 ms (69.75× faster).

Q1. Machine and peak calculations

The machine is a MacBook Air (MacBookAir10,1), with an Apple M1 and 16 GiB unified memory. Native `sysctl` confirms four performance (P) cores and four efficiency (E) cores, eight physical/logical cores and no SMT. All timings are native arm64 on macOS 15.1.1, using Homebrew Clang 23.1.5 with `-O3 -mcpu=apple-m1 -fopenmp-simd`; only the OpenMP variant enables parallel OpenMP regions with `-fopenmp`.

Property	Value / evidence
Core types	Firestorm P + Icestorm E; 4 of each (native probe; Apple [1])
SIMD / registers	128-bit NEON, 4 FP32 lanes; 32 architectural vector registers
FMLA throughput	P: 4/cycle; E: 2/cycle (instruction microbenchmarks [2])
Clock reference	P: 3.204 GHz; E: 2.064 GHz, platform operating points [3]
Base / sustained clock	DVFS; collected sysctl does not expose run-time core clocks
L1D / L1I per P core	128 KiB / 192 KiB (native perflevel0 sysctl)
L1D / L1I per E core	64 KiB / 128 KiB (native perflevel1 sysctl)
L2 clusters	12 MiB shared by 4 P cores; 4 MiB shared by 4 E cores
L3 / cache line	Conventional L3 not reported; system cache unprobed / 128 B
Memory	16 GiB unified; LPDDR4X-4266 platform reference [4]

An FMA counts as two operations. Using the platform maximum operating points, the reference peaks are

$$P_P = 3.204 \times 4 \times 4 \times 2 = 102.528 \text{ GFLOP/s},$$
$$P_E = 2.064 \times 2 \times 4 \times 2 = 33.024 \text{ GFLOP/s},$$
$$P_{all} = 4P_P + 4P_E = 542.208 \text{ GFLOP/s}.$$

These are conditional architectural upper bounds, *not measured sustained clocks*. The P operating point appears as a turbo entry in the Linux/Asahi platform description; its Linux availability does not establish which macOS frequency was used. A fixed base/all-core frequency cannot be reported from this run. Percent-of-peak uses P_P for serial variants; OpenMP uses $P(t) = \min(t, 4)P_P + \max(0, t-4)P_E$, a P-first reference model, not verified core placement.

For the platform memory model, use a 128-bit aggregate interface (two 64-bit-equivalent widths): $B_{ref} = 4266 \times 16/1000 = 68.256$ GB/s. Memory type/rate come from platform identification [4]; the bus width is an explicit model assumption, not a probe of physical LPDDR subchannels. A 384 MiB triad measures useful bandwidth as $B = 12n/t$ (two reads plus one FP32 write), excluding cached-store write allocation. The one-thread rate is 50.552 GB/s; the best is 52.449 GB/s at two threads; eight threads give 42.991 GB/s. Using measured bandwidth, the reference ridge points are $P_P/B_1 = 2.028$ and $P_{all}/B_8 = 12.612$ FLOP/B. With theoretical B_{ref}, they are 1.502 and 7.944 FLOP/B.

Q2. Naive baseline

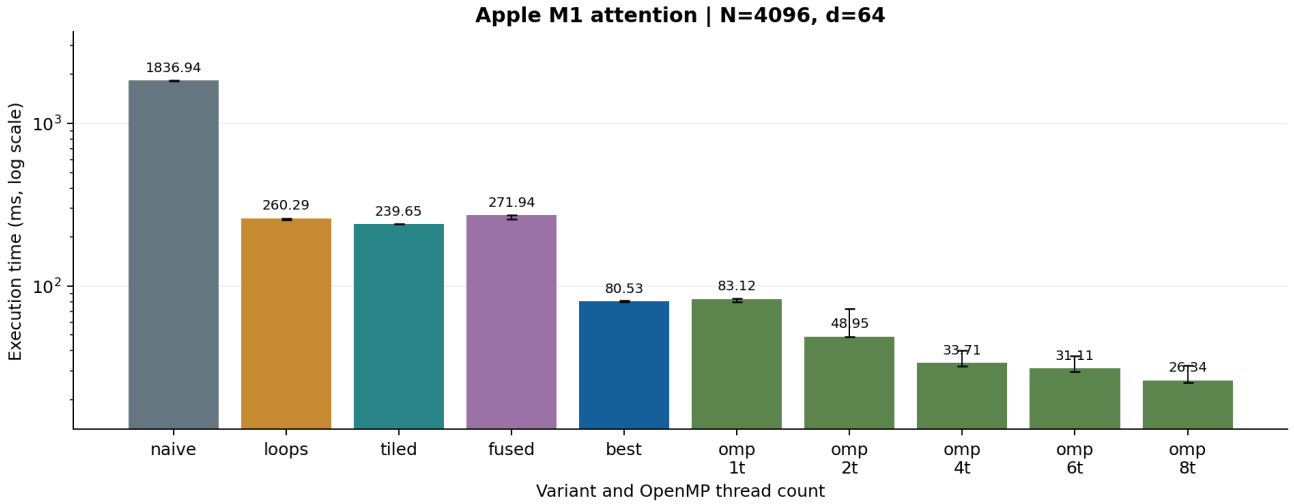
Variant	N	d	Time (ms)	GFLOP/s	% peak*	vs naive
naive	1024	64	109.357	2.45	2.39	1.00×
naive	2048	64	451.060	2.38	2.32	1.00×
naive	4096	64	1836.944	2.34	2.28	1.00×
naive	4096	128	3947.960	2.18	2.12	1.00×

The common work count is $F = 4N^2d$: $2N^2d$ for QK^T and $2N^2d$ for the value product. Reported throughput is $F/(10^9 t)$. Softmax, division, key transposition and allocation remain in the time but not the useful-work numerator, following the starter driver.

At $d = 64$, doubling N multiplies time by 4.12 and 4.07, close to the quadratic prediction of 4. At $N = 4096$, doubling d multiplies time by 2.15: the products double in work, while softmax still handles N^2 scores. Different cache/stride behaviour also matters.

The baseline computes each score as a separate dot reduction. Strict FP32 summation leaves a dependent addition chain. Its value product uses i, k, j loops, so $V[jd + k]$ is read with a 256 B stride at $d = 64$ and 512 B at $d = 128$. The same probability row is traversed for every output feature. This wastes locality and exposes reductions rather than independent contiguous SIMD updates. These are code-based explanations; no per-stage timings or hardware miss counters were collected on the Mac, so exact shares are not claimed.

Protocol. Every original starter driver is preserved. Each invocation performs one warm-up and takes the median of five timed calls; the CSV is the median of three independently warmed invocations per configuration. Only `attention()` is timed; file I/O is excluded, while all temporary allocations and packing are included. Configurations run sequentially. macOS owns P/E placement; no CPU affinity or fixed core frequency is claimed. Three-trial ranges on the time plots are variability indicators, not confidence intervals.



The full CSV contains every input/variant/thread configuration. All four required per-input time plots are supplied. One noisy invocation cannot change a row to its best-case time: the report consistently uses the three-invocation median.

* Peak percentages refer to the Q1 reference architecture/frequency model. GB and GFLOP use decimal units; KiB/MiB/GiB use binary units.

Q3. Single-threaded variants

Variant	N	d	Time (ms)	GFLOP/s	% peak*	vs naive
loops	1024	64	15.915	16.87	16.45	6.87×
loops	2048	64	63.403	16.94	16.52	7.11×
loops	4096	64	260.289	16.50	16.09	7.06×
loops	4096	128	419.289	20.49	19.98	9.42×
tiled	1024	64	14.370	18.68	18.22	7.61×
tiled	2048	64	60.017	17.89	17.45	7.52×
tiled	4096	64	239.650	17.92	17.48	7.67×
tiled	4096	128	422.245	20.34	19.84	9.35×
fused	1024	64	15.761	17.03	16.61	6.94×
fused	2048	64	63.077	17.02	16.60	7.15×
fused	4096	64	271.943	15.79	15.40	6.75×
fused	4096	128	421.148	20.40	19.89	9.37×
best	1024	64	5.175	51.87	50.59	21.13×
best	2048	64	19.304	55.62	54.25	23.37×
best	4096	64	80.528	53.34	52.02	22.81×
best	4096	128	137.706	62.38	60.84	28.67×

(a) Reordered loops. Transpose keys to $T[k, j]$ and use i, k, j score loops. Adjacent key positions become independent SIMD lanes. For SV, use i, j, k so each probability updates contiguous output features from a contiguous row of V. This removes strided column access, but repeatedly updates score/output accumulators and retains the full score matrix.

(b) Cache/register tiling. Use 32×128 outer query/key tiles around 4×32 local accumulator tiles. Four query rows reuse each key/value vector; complete reductions are kept in accumulators before storing results. It improves the 4096×64 case, but is slightly slower than loops at 4096×128 (422.245 vs 419.289 ms). Tiling therefore has shape-dependent benefits; local-array/control overhead can offset cache reuse. The difference is small and is not evidence of a fundamental regression without more trials.

(c) Row fusion. Compute one score row, stable softmax, and its output immediately. At $N = 4096$, score storage falls from 64 MiB to 16 KiB. This version is still slower than tiled at $d = 64$: per-score dot reductions and lost multi-query reuse offset the storage saving. At $d = 128$, loops/tiled/fused are all around 419–422 ms, rather than monotonically improving.

(d) Best M1 implementation. Fuse blocks of 16 query rows; each NEON microkernel handles four query rows by 16 keys/features, using sixteen independent 128-bit accumulators. Reuse each loaded key/value vector across four queries and keep partial sums in registers. A custom vector exponential accelerates stable softmax. At $N = 4096$, each block needs 256 KiB scratch, alongside 1 MiB of transposed keys and 1 MiB of values ($d = 64$), fitting the P-cluster L2. Packing is still timed. Scalar tails handle odd dimensions; the AArch64 fast path was actually exercised in the submitted native Mac run. This is the best implemented/tested variant, without a claim of global optimality or a measured M1 block-size search.

The remaining limits include FMA/load/broadcast throughput, softmax, packing, cache traffic and OS scheduling. Exact bottleneck shares would require profiling beyond this run.

Q4. OpenMP

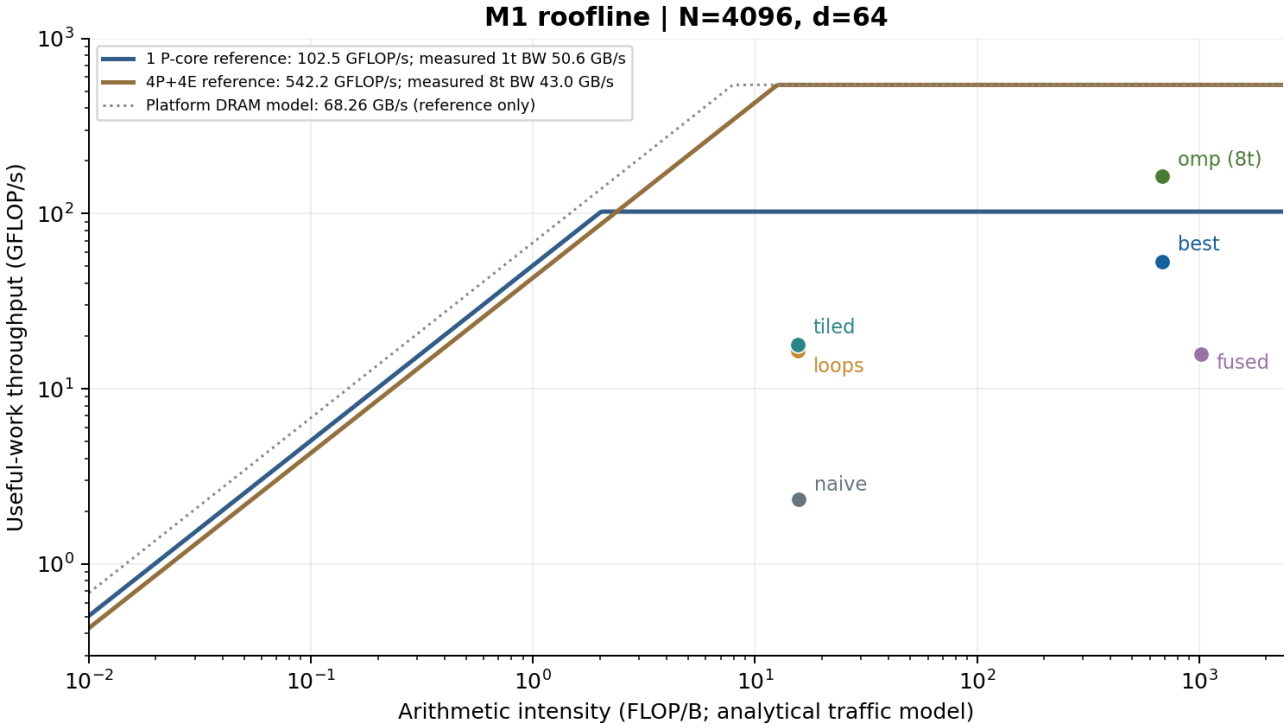
The same compute kernel is used in `best` and `omp`. Key transposition is parallelised over two-dimensional tiles. Query blocks use `schedule(dynamic,1)`: workers own private scratch and disjoint output rows, and read shared keys/values. Dynamic block assignment lets faster workers take more work on the asymmetric 4P+4E CPU. There is no cross-thread output reduction.

The controller disables dynamic team resizing and core binding. Thread placement remains OS-managed. Define $S_t = T_{omp,1}/T_{omp,t}$ and $E_t = S_t/t$, using the one-thread OpenMP implementation as the scaling baseline rather than the separately compiled serial binary.

N	d	Threads	Time (ms)	GFLOP/s	S_t	E_t (%)
1024	64	1	4.832	55.55	1.000	100.0
1024	64	2	2.668	100.61	1.811	90.6
1024	64	4	1.838	146.05	2.629	65.7
1024	64	6	1.615	166.21	2.992	49.9
1024	64	8	1.592	168.62	3.035	37.9
2048	64	1	19.073	56.30	1.000	100.0
2048	64	2	11.031	97.34	1.729	86.5
2048	64	4	8.239	130.32	2.315	57.9
2048	64	6	9.520	112.79	2.003	33.4
2048	64	8	5.773	185.99	3.304	41.3
4096	64	1	83.119	51.67	1.000	100.0
4096	64	2	48.953	87.74	1.698	84.9
4096	64	4	33.710	127.41	2.466	61.6
4096	64	6	31.112	138.05	2.672	44.5
4096	64	8	26.335	163.09	3.156	39.5
4096	128	1	155.675	55.18	1.000	100.0
4096	128	2	85.504	100.46	1.821	91.0
4096	128	4	60.493	142.00	2.573	64.3
4096	128	6	64.284	133.62	2.422	40.4
4096	128	8	57.451	149.52	2.710	33.9

For $N = 4096, d = 64$, eight threads yield 3.156× scaling and 39.5% efficiency, or 69.75× over naive. Eight identical full-speed cores would suggest an 8× line, but this CPU has four slower E cores, shared caches/memory and scheduling overhead. That line is context, not an achievable P-core-only target. Six threads are slower than four for 2048×64 and 4096×128; eight are fastest for all four supplied shapes. The measurements show non-monotonic scaling, not which thread was on which core. Contention, dynamic scheduling, placement and thermal/DVFS effects are plausible contributors, not separately established causes.

Q5. Roofline



Analytical traffic model. Start with $16Nd$ bytes for reading Q, K, V and writing O. For materialised scores, add $16N^2$ bytes across stages: score write, softmax read/write, then value-product read. Softmax's internal row passes are assumed to hit cache. A transposed key buffer adds a conservative $8Nd$ bytes for its write/read. Fused row/block scratch is assumed cache-resident. Ignore write allocation and additional evictions. Hence these idealised intensities are not measured DRAM traffic and may overestimate actual intensity.

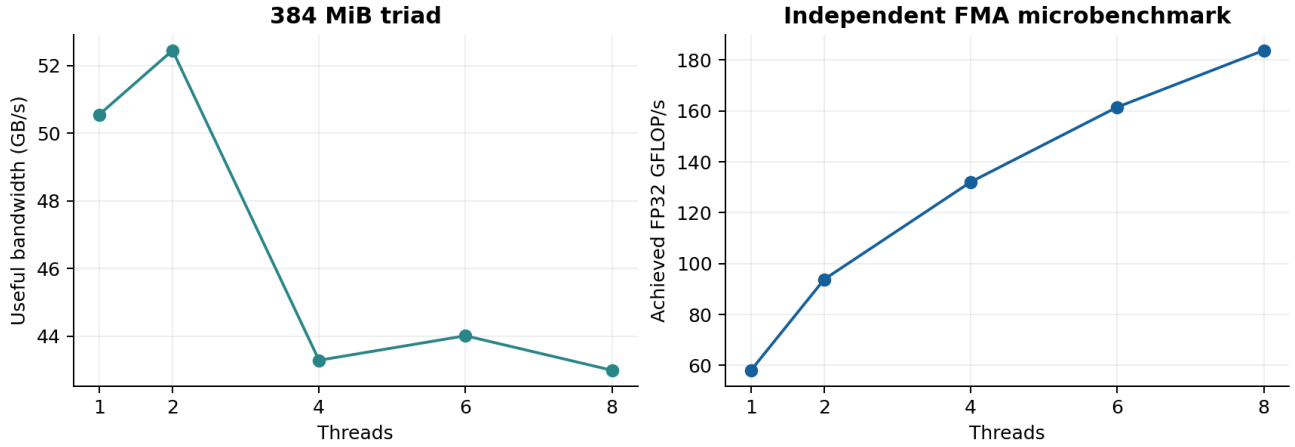
Variant	Modelled bytes	$I = F/B$ (FLOP/B)	GFLOP/s
naive	$16N^2 + 16Nd$	15.754	2.34
loops	$16N^2 + 24Nd$	15.634	16.50
tiled	$16N^2 + 24Nd$	15.634	17.92
fused	$16Nd$	1024.000	15.79
best	$24Nd$	682.667	53.34
omp (8t)	$24Nd$	682.667	163.09

The rooflines use $\min(BI, P)$ with Q1's reference peak and either measured sustained bandwidth or the explicitly assumed platform memory model. Serial points belong against the one-P-core curve; the OpenMP point uses the 4P+4E curve. All points are beyond the corresponding ridge. Thus modelled DRAM bandwidth is not the tight ceiling; this does not prove the kernels are purely compute-bound. Naive suffers reduction/stride/cache-level costs; loops exposes SIMD but repeats accumulator traffic; tiled reuses accumulators. Fused improves storage without equivalent multi-row arithmetic reuse. Best adds explicit SIMD/register tiling and faster softmax. OpenMP raises throughput but does not saturate the aggregate reference peak.

Validation, auxiliary measurements and reproduction

Validation. All 36 non-reference input/configuration rows pass the required $\max|O - O_{naive}|/\max|O_{naive}| \leq 10^{-4}$, including all 20 OpenMP configurations. The worst supplied-input error is 4.73e-06. All 72 independent FP64 checks pass, with a worst error of 4.16e-06; these include odd shapes, tails, zero/uniform logits and large logits. SHA-256 matches all returned C/header sources and four original inputs to the packaged files. The original naive source and all starter drivers are preserved. `make check` rejects missing outputs, invalid headers/lengths, non-finite values and excessive error.

Vector softmax approximation. Subtract the row maximum, then reduce $x = n\ln 2 + r$ with $n = \mathrm{round}(x\log_2 e)$ and $|r| \leq \ln 2/2$. Evaluate $e^x \approx 2^n \sum_{k=0}^{7} r^k/k!$ with NEON FMAs. Values below -87 are set to zero (each contribution below 1.65×10^{-38}); scalar tails use `expf`. The first omitted polynomial term is below 5.2×10^{-9}, but final accuracy also includes FP32 range-reduction, FMA and summation rounding. Output checks establish tolerance on the tested cases, not exactness.



Threads	Triad GB/s	Achieved FMA GFLOP/s
1	50.552	58.076
2	52.449	93.812
4	43.286	132.074
6	44.011	161.481
8	42.991	183.784

The FMA test runs sixteen independent NEON chains, one warm-up and five measurements, counting $16 \times 4 \times 2$ FLOPs per iteration per thread. It reports achieved rates, not a hard roof: one attention shape exceeds its one-thread rate because instruction mix, timing and operating conditions differ. The triad's best bandwidth occurs at two threads; more workers add contention and scheduling costs rather than increasing the measured useful byte rate. No cache-miss, power or sustained-frequency counters were collected.

Reproduce. On the M1, install Homebrew LLVM/libomp/Python, then run `bash src/run_macos.sh`. It compiles the kernels, runs all configurations/checks and collects a result ZIP. `make all; make check` gives the starter-style workflow. `python3 src/finalize_mac.py` rebuilds this PDF, CSV/specification and plots from the included native evidence (requires Matplotlib, NumPy and pdflatex). Use `--data PATH` to supply another complete native run. Regeneration validates source hashes and raw medians.

References and run trace

[1] Apple, MacBook Air (M1, 2020) technical specifications.

https://support.apple.com/en-us/111883

[2a] Dougall Johnson, Apple Microarchitecture Research: Firestorm SIMD/FP, FMLA vector 4S (0.25 cycles per instruction).

https://dougallj.github.io/applecpu/firestorm-simd.html

[2b] Dougall Johnson, Icestorm SIMD/FP, FMLA vector 4S (0.5 cycles per instruction). These are primary reverse-engineering measurements, not Apple specifications.

https://dougallj.github.io/applecpu/icestorm-simd.html

[3] Asahi Linux contributors, Linux T8103 device tree: 3.204 GHz P turbo / 2.064 GHz E operating points; these are reference values, not observed macOS clocks.

https://github.com/torvalds/linux/blob/master/arch/arm64/boot/dts/apple/t8103.dtsi

[4] iFixit, M1 MacBook Pro and Air Teardowns: primary chip identification of LPDDR4X-4266. The report explicitly assumes a 128-bit aggregate interface for the theoretical bandwidth model.

https://www.ifixit.com/News/46884/m1-macbook-teardowns-something-old-something-new

[5] Apple, Tuning your code's performance for Apple silicon: asymmetric cores and scheduling.

https://developer.apple.com/documentation/apple-silicon/tuning-your-code-s-performance-for-apple-silicon/

[6] Assignment starter, commit 29f5b9bf4aa89f74f425b1797cc46cdb457f5569.

https://github.com/NMJ0/Assignment-5-Systems-for-Deep-learning